Statistical Methods in Education Research
Vanderbilt University
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46 total points

Problem Set 6 SOLUTIONS

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1. (**12 points–2 each**) Suppose that the historical mean number of days absent among high school students in a state is 8 days per year, with a population standard deviation of 6 days. A researcher suspects that absenteeism has increased. She selects a random sample of records for 400 students and finds that the average number of days absent in the sample is 8.6. Assume that the population standard deviation remains 6 days.

   (a) Write down H_0 and H_1 in terms of the population mean μ.

   (b) Describe the approximate sampling distribution of $\bar{x}$ if H_0 is true. What are its mean and standard error?

   (c) Calculate the appropriate test statistic.

   (d) Calculate the p-value. Interpret this in words.

   (e) Test the hypothesis at significance levels $\alpha = 0.05$ and $\alpha = 0.01$. What do you conclude?

   (f) A school board member interprets your findings as follows: "We have demonstrated that student absenteeism has increased." Is this an appropriate interpretation? Explain.

   (a) Let μ represent the population mean number of days absent per year among high school students.

   $$H_0 : \mu = 8 \qquad H_1 : \mu > 8$$

   (b) If H_0 is true, the sampling distribution of $\bar{x}$ is approximately normal, with mean $\mu_0 = 8$ and standard error

   $$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{400}} = 0.30.$$

   (c) The test statistic is

   $$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{8.6 - 8}{0.30} = 2.00.$$

   The observed sample mean is 2 standard errors above the hypothesized pop-

ulation mean.

(d) The one-sided p-value is

$$p = \Pr(Z \geq 2.00) \approx 0.0228.$$

In Stata: `display 1-normal(2)`

If the true population mean were 8 days, the probability of obtaining a sample mean of 8.6 days or higher in a random sample of 400 students would be approximately 2.28%.

(e) At $\alpha = 0.05$, we reject H_0 because $p = 0.0228 < 0.05$. There is statistically significant evidence that mean absenteeism exceeds 8 days.

At $\alpha = 0.01$, we do not reject H_0 because $p = 0.0228 > 0.01$. The evidence is not sufficiently strong to reject the null at the 1% level.

(f) The school board member's statement is too strong. We have found statistically significant evidence that mean absenteeism exceeds 8 days at the 5% level, but we have not *proved* that absenteeism has increased.

It remains possible that the population mean is 8 and that our sample mean was unusually high due to sampling variability.

2. (**6 points**) A researcher tests whether average daily screen time among adolescents exceeds 4 hours per day. She obtains a p-value of 0.032. Consider the following statements. For each, indicate whether it is correct or incorrect, and briefly explain your answer.

(a) There is a 3.2% probability that the null hypothesis is true.

(b) If the null hypothesis is true, the probability of obtaining a test statistic at least as extreme in the direction of the alternative as the one observed is 3.2%.

(c) The researcher should reject the null hypothesis at the 5% significance level.

(d) The researcher should reject the null hypothesis at the 1% significance level.

(e) The researcher has demonstrated that average daily screen time exceeds 4 hours.

(f) If the researcher does not reject the null hypothesis, she has established that average screen time is 4 hours.

(a) **Incorrect.** The p-value is not the probability that H_0 is true. It is calculated under the assumption that H_0 is true.

(b) **Correct.** The p-value of 0.032 means that, if H_0 is true, there is a 3.2% probability of obtaining a test statistic at least as extreme as the observed statistic in the direction of H_1.

(c) **Correct.** Since $p = 0.032 < 0.05$, we reject H_0 at the 5% significance level.

(d) **Incorrect.** Since $p = 0.032 > 0.01$, we do not reject H_0 at the 1% significance level.

(e) **Incorrect.** Although we reject H_0 at the 5% level, this does not prove that mean screen time exceeds 4 hours. There remains a possibility that H_0 is true and the sample produced an unusually high estimate.

(f) **Incorrect.** Failure to reject H_0 does not establish that H_0 is true. It simply means that the observed data do not provide sufficient evidence to reject H_0 at the chosen significance level.

3. (**8 points–2 each**) Suppose the average reading score among students in a particular state has historically been 500 points, with a known population standard deviation of 80 points. A researcher believes that the mean has declined and plans to test:

$$H_0 : \mu = 500 \qquad H_1 : \mu < 500$$

She collects a random sample of 256 students.

(a) What is the standard error of the sample mean?

(b) At a significance level of $\alpha = 0.05$, what is the largest value of $\bar{x}$ for which the researcher would reject H_0?

(c) Suppose the sample mean is 492. Would you reject H_0 at the 5% level? Explain.

(d) If the researcher instead uses $\alpha = 0.01$, would the critical value (as you found in part b) be higher or lower? Explain intuitively before calculating it.

(a) The standard error of the sample mean is

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{80}{\sqrt{256}} = 5.$$

(b) For a lower-tailed test with $\alpha = 0.05$, the critical value of the standard normal distribution is $z = -1.645$. We reject H_0 when

$$\frac{\bar{x} - 500}{5} < -1.645.$$

Solving for $\bar{x}$:

$$\bar{x} < 500 - 1.645(5) = 491.775.$$

Thus, the critical value of the sample mean is approximately 491.78. We reject H_0 when the sample mean falls below this value.

(c) If $\bar{x} = 492$, the test statistic is

$$z = \frac{492 - 500}{5} = -1.60.$$

The one-sided p-value is

$$p = \Pr(Z \leq -1.60) \approx 0.0548.$$

In Stata: `display normal(-1.6)`

Since $p > 0.05$, we do not reject H_0.

Equivalently, 492 exceeds the critical sample mean of 491.78 and therefore falls outside the rejection region.

(d) The critical value of $\bar{x}$ would be *lower* if we used $\alpha = 0.01$.

A smaller significance level requires stronger evidence against H_0. For this lower-tailed test, the sample mean must be even further below 500 before we reject the null hypothesis.

The critical standard normal value is approximately $z = -2.326$. Thus,

$$\bar{x}_{crit} = 500 - 2.326(5) \approx 488.37.$$

We would reject H_0 only for sample means below approximately 488.37.

4. (**8 points–2 each**) An education researcher evaluates a new instructional program and reports that students exposed to the program have a higher average test score than a historical benchmark. The researcher obtains a one-sided p-value of 0.048.

   (a) Would the researcher reject H_0 at $\alpha = 0.10$, $\alpha = 0.05$, and $\alpha = 0.01$?

   (b) Suppose the p-value had instead been 0.052. How would the conclusions change?

   (c) Is there a meaningful difference in the strength of the statistical evidence between a p-value of 0.048 and a p-value of 0.052?

   (d) A news article reports that the first study "proved the program works" while the second study "found no evidence that the program works." Critique these characterizations.

   (a) With a p-value of 0.048:

   - At $\alpha = 0.10$: Reject H_0.
   - At $\alpha = 0.05$: Reject H_0.
   - At $\alpha = 0.01$: Do not reject H_0.

   (b) With a p-value of 0.052:

   - At $\alpha = 0.10$: Reject H_0.
   - At $\alpha = 0.05$: Do not reject H_0.
   - At $\alpha = 0.01$: Do not reject H_0.

   Thus, the decision changes at the 5% level, but not at the 10% or 1% levels.

   (c) No. The difference between $p = 0.048$ and $p = 0.052$ is very small. Although the two values produce different decisions when using $\alpha = 0.05$, they represent nearly identical strengths of evidence against H_0. The 5% threshold is a convention rather than a point at which the evidence changes abruptly.

   (d) Both characterizations are misleading. The first study did not *prove* that the program works. It found statistically significant evidence against H_0 at the 5% level.

   The second study did not establish that the program is ineffective, nor did it find an absence of evidence. Its p-value of 0.052 indicates nearly the same strength of evidence against H_0 as the first study.

   A more informative comparison would consider the estimated effects, standard errors, confidence intervals, and p-values rather than relying exclusively on a binary significance threshold.

5. (**12 points–2 each**) Two researchers are independently investigating whether mean mathematics achievement has increased above a historical average of 500. Both researchers obtain a sample mean of 505. Both assume that the population standard deviation is $\sigma = 100$. However, they use different sample sizes.

	Researcher A	Researcher B
Sample mean	505	505
Population SD	100	100
Sample size	100	2,500

   (a) Calculate the standard error of the sample mean for each researcher.

   (b) Calculate the test statistic and one-sided p-value for each researcher.

   (c) At $\alpha = 0.05$, which researcher rejects $H_0 : \mu = 500$?

   (d) Both researchers estimate the same difference of 5 points. Why do they reach different conclusions?

   (e) Does a statistically significant finding necessarily indicate that the estimated difference is substantively important for education policy? Explain, and speculate on how you might determine whether a difference is "substantively important."

   (f) Calculate 95% confidence intervals for the population mean using each sample. Compare their widths.

   (a) The standard errors are

$$SE_A = \frac{100}{\sqrt{100}} = 10$$

   and

$$SE_B = \frac{100}{\sqrt{2500}} = 2.$$

   Researcher B has a smaller standard error because the sample size is larger.

   (b) Both researchers test

$$H_0 : \mu = 500 \qquad H_1 : \mu > 500.$$

   For Researcher A:

$$z_A = \frac{505 - 500}{10} = 0.50$$

$$p_A = \Pr(Z \geq 0.50) \approx 0.3085.$$

For Researcher B:

$$z_B = \frac{505 - 500}{2} = 2.50$$

$$p_B = \Pr(Z \geq 2.50) \approx 0.0062.$$

In Stata:

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display 1-normal(0.5)
display 1-normal(2.5)
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(c) At $\alpha = 0.05$:
Researcher A does not reject H_0, since $p_A = 0.3085 > 0.05$.
Researcher B rejects H_0, since $p_B = 0.0062 < 0.05$.

(d) The researchers obtain the same point estimate of μ and the same estimated difference of 5 points above the hypothesized mean.

However, Researcher B has a much larger sample, which produces a smaller standard error.

Consequently, the same estimated difference is further from the hypothesized mean in units of standard errors, yielding a larger test statistic and smaller p-value.

(e) No. Statistical significance indicates that the observed result would be unlikely under the null hypothesis. It does not necessarily indicate that the estimated difference is large or substantively important.

For example, a 5-point increase in test scores might be statistically significant in a large sample but relatively small in terms of its educational importance. Lecture 7 will cover practical significance in more detail, but there are a number of ways to assess this. One is two compare the difference to the overall standard deviation in the outcome (a measure of effect size). Another is to compare the difference to relevant benchmarks in the field (e.g., a "week of learning").

(f) The 95% confidence interval is

$$\bar{x} \pm 1.96 \left(\frac{\sigma}{\sqrt{n}} \right).$$

For Researcher A:

$$505 \pm 1.96(10)$$

$$CI_A = (485.40,\ 524.60).$$

For Researcher B:

$$505 \pm 1.96(2)$$

$$CI_B = (501.08,\ 508.92).$$

Researcher B's confidence interval is considerably narrower because the sample mean is estimated more precisely.

Note that the interval for Researcher A includes 500, while the interval for Researcher B does not.